QIU, XINRU

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Professional Summary

Computational biologist and AI researcher specializing in single-cell genomics, foundation models, causal inference, and therapeutic target discovery. Builds reproducible machine-learning frameworks and analytical pipelines for biomarker discovery, foundation-model evaluation, and causal therapeutic-target prioritization across cancer, immunology, and translational medicine. First-author work spans single-cell sepsis dynamics, AI for protein structure prediction, platelet machine learning, and causal target-discovery agents.

Selected Impact

- 20+ peer-reviewed publications with 400+ citations and an h-index of 10 (Google Scholar)
- First-author publications cited 115+, 101+, and 31+ times respectively
- Completed over 200 verified peer reviews across 130+ manuscripts for leading journals and conferences
- Reviewer for ICLR, ICML, NeurIPS, AI4Science, and GEM workshops; grant reviewer for UCR undergraduate research awards
- Contributor to an NIH R01 and author of the R01 proposals
- Developed AI and single-cell genomics frameworks for biomarker discovery and therapeutic target prioritization, released as open-source software (`scfm-benchmark`, CausalTrapBench, CCTL) with documentation, Docker, and CI

Research Metrics

Google Scholar Citations: 400+ | **h-index:** 10 | **Peer-Reviewed Publications:** 20+ | **First-Author (high-impact):** 3 | **Peer Reviews Completed:** 200+ | **Manuscripts Reviewed:** 130+ | **Conference Reviews:** ICLR, ICML

Core Technical Skills

Languages: Python, R, Bash, SQL, Git | **ML:** PyTorch, TensorFlow, scikit-learn, XGBoost, Hugging Face | **Single-Cell:** Seurat, Scanpy, CellRanger, Monocle3 | **Workflows:** Nextflow, Snakemake, WDL/Cromwell | **Compute:** HPC/SLURM, GPU/CUDA, Docker, Singularity, Google Cloud Platform

Education

PhD University of California, Riverside (UCR) – Riverside, CA June 2023
Genetics, Genomics, and Bioinformatics (GGB)
Committee: Adam Godzik (chair), Meera Nair, Zhenyu Jia

MS University of Southern California (USC) – Los Angeles, CA May 2015
Pharmaceutical Sciences
Advisor: Wei-Chiang Shen

BS Jilin University (JLU) – Changchun, China June 2013
Pharmacy

Professional Experience

University of California, Riverside – Riverside, CA July 2023 – Present
Postdoctoral Researcher, Advisor: Adam Godzik

- **Single-Cell Foundation Model Robustness Benchmark (Targeting *Briefings in Bioinformatics*):** Designed and led an out-of-distribution robustness benchmark of five single-cell foundation models (STATE, UCE, Geneformer, scPRINT-2, scGPT) for in-hospital mortality prediction from human platelet transcriptomes. Built a canonical-input pipeline feeding identical raw counts to each model for apples-to-apples comparison, and implemented patient-level StratifiedGroupKFold cross-validation with donor-clustered inference to eliminate the cell-level leakage common in published FM evaluations. Key findings: a simple PCA + logistic-regression baseline ties or beats the foundation models on most axes (adaptation-curve asymptote 0.96); only STATE generalizes bidirectionally across diseases; scGPT is anti-correlated on cross-disease transfer (reported as a deliberate negative result); and LoRA fine-tuning yields only marginal gains over zero-shot embeddings — challenging the field's default "fine-tune for downstream" assumption. Released `scfm-benchmark` as an open-source PyPI package with CLI, documentation, Docker, and CI — a drop-in robustness audit for any new model. Sole first author.

- **ML-Based Regulatory Network Interpretation for Therapeutic Target Discovery in Obesity Immunology:** Led core computational analyses for an NIH-funded systems biology project (co-first author), designing a complete single-cell RNA-seq framework for 97,061 immune cells across 16 multiplexed samples (10x Genomics FLEX) and 6 experimental conditions. Identified 6 transcriptionally distinct adipose eosinophil subtypes and translated gradient-boosting regulatory-network predictions (GRNBoost2/SCENIC) into prioritized, subtype-specific transcription-factor hypotheses via a composite specificity/variance scoring framework. Reconstructed differentiation dynamics with multi-method trajectory inference (CellRank2, scVelo, Monocle3) and mapped ligand-receptor-TF signaling with CellChat, identifying CCL6-CCR3/CCR2 as a sex-dimorphic checkpoint of adipose inflammation. Applied rigorous statistics (pseudobulk DESeq2 differential expression, Bayesian compositional analysis with `sccomp`, DecontX ambient-RNA correction). *Co-first author manuscript in revision (Ruggiero-Ruff*, Qiu* et al. 2026).*
- **The Oracle Gap — A Causal-Inference Diagnostic and Benchmark for Single-Cell Target Discovery:** Built a causal-inference *diagnostic* for target discovery — an instrument that measures how much causal signal a ranker leaves unrealized, not a new ranking method. Defined the *oracle gap*, a metric-agnostic gap between a learned ranker and an oracle holding the true causal graph, and **CausalTrapBench**, a benchmark of paired true-parent / verified-decoy / oracle-graph tasks that scores decoy suppression rather than graph recovery, so rankers that latch onto confounders or mediators are penalized. The diagnostic's core rests on generic off-the-shelf rankers (correlation, Lasso, NOTEARS, plus causal-discovery learners PC/GES/IAMB/DirectLiNGAM): a method-agnostic sensitivity analysis shows the achievable ceiling is set by an *identification* limit (confounding strength, irreducible with more data) rather than an *estimation* limit (sample size), and that the dominant nonlinear failure mode is governed by the conditional-independence test, not the search family. Demonstrated the diagnostic on genome-wide CRISPR/Perturb-seq data ($\sim$2M cells) using a two-pass contrastive reference ranker (CCTL) as a worked case study — it narrows the cross-dataset median rank of the true perturbed gene from 72 to 8, with the gap analysis quantifying the residual causal ceiling. Shipped fully reproducible artifacts: a claim-traceability ledger, an anonymized code/data supplement, and installable `causaltrap` and `cctl` packages (Docker, CI). Sole first author.
  Manuscript targeting AAAI 2027 (submission July 2026).
- **Verification-Aware AI Agent for Therapeutic Target Validation from Single-Cell Transcriptomics:** Designed and implemented a six-layer (L1–L6) agentic target-*validation* system ("propose, then verify") that turns a ranked candidate list into auditable per-gene verdicts (`supported/partial/refuted/unknown`). A source-agnostic hypothesis layer (CCTL is one example generator) feeds a deterministic cascading critic — contradiction $\rightarrow$ causal $\rightarrow$ clinical (L3–L5) — governed by a sixteen-rule written constitution (Constitutional AI pattern) that reconciles causal, statistical, and public-database evidence (PubMed, OpenTargets, DGIdb, UniProt, DepMap) and audits agreement before any recommendation. Decoupled the deterministic verifier from the language model so the same critic audits rule-based, LLM-backed, or DSPy-generated reasoning identically; orchestrated by default with a LangGraph `StateGraph` (bit-identical to a pure-Python reference) plus optional mem0 cross-run memory, a DepMap pan-essentiality gate, and AMIE-rubric-scored recommendations.
  Validation (three complementary axes): (i) TPR = TNR = 1.00 on a prospective ten-gene FDA-target vs. housekeeping panel, where a deterministic DepMap pan-essentiality gate rejects the ribosomal gene RPL13 in 100% of runs while an unconstrained LLM recommender would admit it as a therapeutic target in 4/5 runs ($\sim$80%) — a worked example that deterministic gates and LLM reasoning are complementary defenses against different error classes, not interchangeable substitutes; (ii) an external melanoma anti-PD-1 recall@K backtest (Sade-Feldman GSE120575, 16,291 cells) against six FDA-approved checkpoint targets, establishing a reproducible accuracy ceiling for any per-source ranker; and (iii) a retrospective sepsis case study ($\sim$57,000 PBMCs) recovering CD52 at rank #3 of 33,538 with thirteen agreeing evidence lines (in silico perturbation, Mendelian randomization on GTEx + eQTLGen instruments, ENCODE ChIP-seq, survival modeling). Engineered the validation panel to be reproducible *offline* — bit-identical from committed, dated evidence-dossier snapshots (no network) plus a single provenance JSON — and shipped a laptop-runnable CLI. Sole developer of the agent stack.
  Manuscript in preparation (2026).
  Predecessor open-source repository: github.com/xqiu625/ai-therapeutic-target-discovery
- **Image-Anchored Spatial Transcriptomics of Kidney Allograft Rejection (bioimage + spatial):** Contributed

to the first spatial transcriptomic study of kidney allograft rejection from FFPE core needle biopsies using 10X Genomics Visium. Analyzed 8 biopsies across 4 diagnostic groups (non-rejection, C4d+ active AMR, C4d- active AMR, acute TCMR, chronic active AMR); performed spatial clustering and cell type annotation to identify distinct FCGR3A^{hi} monocyte/macrophage subclusters uniquely enriched in acute rejection. Anchored the spatially resolved transcriptomes to tissue histology, mapping immune cell distribution onto pathologist-annotated histopathological lesions (peritubular capillaritis, glomerulitis, tubulitis) and revealed trained immunity metabolic reprogramming (glycolysis, oxidative phosphorylation) as a mechanism in transplant rejection. Distinguished C4d+ from C4d- AMR transcriptomically, which bulk microarray approaches could not achieve.
Related publication: Frontiers in Immunology (2025); GitHub: github.com/xqiu625/spatial-kidney-rejection

- **Cardiovascular Proteomics & Bioimage Analysis (self-directed case study):** Integrated a public mass-spectrometry platelet proteome with single-cell platelet transcriptomes (RNA–protein concordance, Spearman $\rho = 0.49$ over 6,902 shared genes), and performed imaging-based spatial transcriptomics (10x Xenium) on human carotid atherosclerosis (120,164 segmentation-derived cells across two gene panels): `squidpy` per-section neighborhood enrichment, Ripley's K, anatomical-compartment mapping (fibrous cap / necrotic core), symptomatic-vs-asymptomatic contrast, and cross-panel replication. Extends the platelet research line into cardiovascular **proteomics** and **bioimage / spatial** analysis — building mass-spectrometry-proteome integration and imaging-based spatial capability. *GitHub: github.com/xqiu625/cardiovascular-proteomics-bioimage*
- **AI Applications in Protein Structure Prediction and Drug Discovery:** Published first-author review in *Biomolecules* (Editor's Choice, 70+ citations as of Jan 2026) providing comprehensive analysis of AI-driven structure prediction tools (AlphaFold2, ESMFold, RoseTTAFold, OpenFold) and their applications in cancer drug discovery. Evaluated generative AI approaches for de novo protein design (ProGen, ProteinMPNN, RFdiffusion) and molecular docking (DiffDock), establishing foundational reference for structure-based therapeutic development. *DOI: 10.3390/biom14030339*

University of California, Riverside – Riverside, CA Jan 2019 – June 2023
Ph.D. Research Assistant, Advisor: Adam Godzik

- **Platelet Transcriptional Profiling in Inflammatory Diseases:** Investigated platelet transcriptional heterogeneity in COVID-19, sepsis, and systemic lupus erythematosus using single-cell transcriptomics (47,977 platelets from 413 samples) coupled with machine learning (Deep Neural Network, XGBoost). Identified distinct platelet subpopulations associated with disease severity and clinical outcomes; revealed that platelet transcriptional changes exacerbate endotheliopathy and disseminated intravascular coagulation in fatal cases. Discovered 21 consensus biomarkers distinguishing survival vs. fatal platelets and defined gene modules for therapeutic targeting. *First-author publication: Int. J. Mol. Sci. 2024; DOI: 10.3390/ijms25115941. GitHub: github.com/xqiu625/PlateletSubpop-ML-ScTranscriptomics*
- **Sepsis Biomarker Discovery Through Single-Cell Dynamics:** Characterized dynamic transcriptional changes in human immune cells during fatal sepsis using single-cell RNA sequencing. Implemented comprehensive bioinformatics pipeline (Cell Ranger, Seurat, Scanpy, SingleR, scCATCH, Monocle) to identify cell-type-specific gene expression signatures associated with disease progression and patient outcomes. Discovered novel biomarkers and elucidated key cellular pathways in sepsis pathogenesis, revealing immune cell dysfunction patterns that distinguish fatal from survival outcomes. *Co-first author: J. Leukoc. Biol. 2021; DOI: 10.1002/JLB.5MA0721-825R. GitHub: github.com/xqiu625/sepsis-scrna-dynamics*
- **Cross-Disciplinary Research Collaborations:** Co-authored publications investigating protein structures (TpLRR/BspA-like proteins, *J. Struct. Biol. 2023*), T cell memory responses in Toxoplasma infection, and clinical outcomes in COVID-19 health disparities. Applied computational analysis and machine learning techniques across diverse biological systems, demonstrating versatility in structural bioinformatics, immunology, and clinical research applications.

Illumina – San Diego, CA June 2022 – Sep 2022
Bioinformatics Intern – Doctoral

Developed pharmacogenomics interpretation and clinical-reporting workflows within Illumina's DRAGEN Bio-IT platform, used by clinical laboratories worldwide.

- **Pharmacogenomics Algorithm Development:** Implemented star allele calling algorithms for CYP2D6, CYP2C19, and other pharmacogenes, incorporating copy number variation (CNV) detection and haplotype phasing for accurate diplotype inference. Developed genotype-to-phenotype translation logic following CPIC guidelines for clinical decision support.
- **Knowledge Base Integration & Validation:** Engineered data parsing and standardization pipelines for

PharmVar, PharmGKB, and CPIC databases; implemented clinical evidence level assessment logic and cross-database validation framework to ensure annotation accuracy and regulatory compliance.

- **Clinical Reporting System:** Established end-to-end workflow for automated pharmacogenomics report generation using Illumina DRAGEN Bio-IT platform with TypeScript, Node.js, and HTML, enabling scalable clinical genomics reporting across cross-functional teams.

Vantari Genetics – Irvine, CA Nov 2015 – Sep 2018
Clinical Genomics Scientist
Contributed to development and implementation of Next-Generation Sequencing (NGS) pipelines for hereditary cancer testing, with focus on variant classification and interpretation following ACMG guidelines.

- **NGS Pipeline Development:** Worked closely with Bioinformatics Pipeline Engineer to create integrated workflows for variant discovery and interpretation. Contributed to pipeline validation efforts, ensuring accuracy and reliability of genetic testing results through rigorous quality control procedures.
- **Variant Classification and Interpretation:** Conducted NGS hereditary cancer multigene tests, interpreting complex genetic data using ACMG classification guidelines. Classified genetic variants with consistency and accuracy in reporting. Developed and implemented standardized workflows and SOPs for variant classification, improving efficiency and reducing error rates in clinical genomics testing.
- **Data Analysis and Clinical Reporting:** Analyzed data from key genomic databases including RefSeq, UCSC Genome Browser, ClinVar, HGMD, TCGA, ENCODE, UniProt, GEO, Ensembl, COSMIC, and dbSNP. Prepared concise, clinically relevant patient reports based on genetic findings and current literature for healthcare providers.
- **Training and Knowledge Dissemination:** Organized and conducted training sessions for new hires on NGS technologies and variant interpretation methodologies. Contributed to development of educational materials for both internal use and client education on genetic testing interpretation.

University of Southern California – Los Angeles, CA June 2014 – May 2015
Graduate Research Assistant

- **Peptide Drug Design & Medicinal Chemistry:** Conducted computational and experimental research on pH-sensitive cell-penetrating peptides (CPPs) fused with histidine-glutamate co-oligopeptides for targeted drug delivery applications. Applied rational drug design principles to optimize peptide sequences for enhanced cellular uptake and cytotoxicity, investigating structure-activity relationships (SAR) and drug metabolism considerations. Designed and executed experiments evaluating ADMET properties including cytotoxicity assays, cell permeability studies, and pH-dependent activity profiling, demonstrating understanding of medicinal chemistry techniques and pharmacokinetics principles for peptide therapeutics.
- **Computational Modeling & Data Analysis:** Utilized computational chemistry approaches and molecular modeling to predict peptide conformations, binding interactions, and cellular uptake mechanisms. Performed data analysis using Python and statistical software to assess structure-function relationships, generating detailed scientific reports on peptide optimization strategies. Collaborated with interdisciplinary team members to advance peptide-based therapeutic design, contributing to fundamental understanding of drug delivery systems and targeted therapeutics.

Honors and Awards

- *Biomolecules* (MDPI) Best Paper Award, for “Advances in AI for Protein Structure Prediction: Implications for Cancer Drug Discovery and Development,” 2026
- Higher Education Emergency Relief Funds (HEERF) Dissertation Year Fellowship ($7,200), 2022
- Annual GGB Symposium Best Poster Award, 2022
- Trainee Poster Award, American Association of Immunologists (AAI), 2022
- 4th Annual Ultimate Biomed Retreat Award for "Best Poster", UCR, 2022
- Student Award of Excellence (2,000 RMB, awarded to one student per class), JLU, 2012
- Student Award of Excellence (2,000 RMB, awarded to one student per class), JLU, 2011
- National Fellowship (8,000 RMB), Ministry of Education, China, 2010
- Outstanding Student Award (2,000 RMB, awarded to one student per class), JLU, 2009

Teaching Experience

National Institute of Allergy and Infectious Diseases (NIAID), NIH – Seattle, WA Dec 2024
Workshop on Computational Modeling of Proteins for Infectious Disease

- Led hands-on workshop for infectious disease researchers, providing instruction on computational protein modeling techniques and tools including AlphaFold2 and ESMFold.
- Designed and delivered practical demonstrations of protein structure prediction and analysis tools for biomedical applications and drug discovery.

University of California, Riverside – Riverside, CA Sep 2023 – Dec 2024
Adam Godzik, School of Medicine Division of Biomedical Sciences

- Taught 'BMSC 202 - Molecular Basis of Disease' at Master's level to cohort of 25 students. Course content included in-depth analysis of disease etiology at molecular level, with focus on contemporary discoveries and diverse methodological applications.
- Designed and implemented practical R coding exercises to improve students' programming skills for biomedical data analysis.

University of California, Riverside – Riverside, CA Sep 2021 – Dec 2021
Ayala Rao, Department of Plant Pathology & Microbiology

- Taught 'BIOL107A - Molecular Biology,' undergraduate course with enrollment of 160 students. Topics included genetics, recombinant DNA technology, molecular cloning, and virology.
- Composed and administered quizzes and examinations to assess academic performance and student comprehension.
- Conducted weekly group discussions to enhance collaborative learning and critical thinking skills.
- Held regular office hours and provided supplemental educational materials to support student learning and success.

University of California, Riverside – Riverside, CA Mar 2021 – June 2021
James Burnette, Department of Evolution, Ecology and Organismal Biology

- Taught 'BIOL020- Dynamic Genome' undergraduate course with average attendance of 60 students per quarter, encompassing plant and animal genomics, bioinformatics tools, and genomic databases.
- Constructed and graded quizzes, exams, and homework to evaluate student comprehension of genomic concepts.
- Collaborated with team of four teaching assistants to streamline lab sessions and grading processes.

Undergraduate Students Advised

- Keita Ichii, 2023 Winter, Spring, Fall
- Abraham Takkouche, 2023 Winter, Spring, Fall
- Jacob Sola, 2023 Winter, Spring, Fall
- Jasmine Nguyen, 2022 Spring
- Britney Margheim, 2019 Fall
- Mabel Shehada, 2019 Fall
- Dev Tanna, 2019 Fall

Publications

20+ peer-reviewed publications | 400+ citations | h-index 10 (Google Scholar)

Journal Publications & Preprints

1. **Qiu, X.**, Li, J., Bonenfant, J., Jaroszewski, L., Mittal, A., Klein, W., Godzik, A., & Nair, M. G. (2021). Dynamic changes in human single-cell transcriptional signatures during fatal sepsis. *Journal of Leukocyte Biology, 110*(6), 1253–1268. (Highly Cited 115+)
2. **Qiu, X.**, Li, H., Ver Steeg, G., & Godzik, A. (2024). Advances in AI for protein structure prediction: Implications for cancer drug discovery and development. *Biomolecules, 14*(3), 339. (Highly Cited 101+; *Biomolecules* Best Paper Award)
3. **Qiu, X.**, Nair, M. G., Jaroszewski, L., & Godzik, A. (2024). Deciphering abnormal platelet subpopulations in COVID-19, sepsis and systemic lupus erythematosus through machine learning and single-cell transcriptomics. *International Journal of Molecular Sciences, 25*(11), 5941. (Cited 31+)

4. Koury, J., Singh, H., Sutley-Koury, S. N., Fok, D., **Qiu, X.**, Maung, R., ... Kaul, M. (2025). EphB2-mediated ephrin-B reverse signaling on microglia drives an anti-viral, but inflammatory and neurotoxic response associated with HIV. *Journal of Neuroinflammation, 22*(1), 1–18.
5. Chen Wongworawat, Y., Nepal, C., Duhon, M., Chen, W., ... **Qiu, X.**, & Wang, C. (2025). Spatial transcriptomics reveals distinct role of monocytes/macrophages with high FCGR3A expression in kidney transplant rejections. *Frontiers in Immunology, 16*, 1654741.
6. He, R., Sarwal, V., **Qiu, X.**, Zhuang, Y., Zhang, L., Liu, Y., & Chiang, J. (2025). Generative AI models in time-varying biomedical data: Scoping review. *Journal of Medical Internet Research, 27*, e59792.
7. Harahap-Carrillo, I. S., Fok, D., Wong, F., Malik, G., Maung, R., **Qiu, X.**, Ojeda-Juárez, D., ... Kaul, M. (2025). Chronic, low-dose methamphetamine reveals sexual dimorphism of memory performance, histopathology, and gene expression affected by HIV-1 Tat protein in a transgenic model of NeuroHIV. *Viruses, 17*(3), 361.
8. Yisrael-Gayle, K., Helderop, E., Morin, C., Polk, D., **Qiu, X.**, Dingilian, H., ... & Lo, D. D. (2025). Evidence for Aerosolized Environmental Bacterial Endotoxin as an Environmental Health Hazard. *medRxiv*, 2025-10.
9. Vizcarra, E. A., Ulu, A., Landrith, T. A., **Qiu, X.**, Godzik, A., & Wilson, E. H. (2023). Group 1 metabotropic glutamate receptor expression defines a T cell memory population during chronic *Toxoplasma* infection that enhances IFN-gamma and perforin production in the CNS. *Brain, Behavior, and Immunity, 114*, 131–143.
10. Takkouche, A., **Qiu, X.**, Sedova, M., Jaroszewski, L., & Godzik, A. (2023). Unusual structural and functional features of TpLRR/BspA-like LRR proteins. *Journal of Structural Biology, 215*(3), 108011.
11. Li, J., Ruggiero-Ruff, R. E., He, Y., **Qiu, X.**, Lainez, N., Villa, P., Godzik, A., Coss, D., & Nair, M. G. (2023). Sexual dimorphism in obesity is governed by RELMα regulation of adipose macrophages and eosinophils. *eLife, 12*, e86001.
12. Bergersen, K. V., Pham, K., Li, J., Ulrich, M. T., Merrill, P., He, Y., Alaama, S., **Qiu, X.**, Harahap-Carrillo, I. S., Ichii, K., Frost, S., Kaul, M., Godzik, A., Heinrich, E. C., & Nair, M. G. (2023). Health disparities in COVID-19: Immune and vascular changes are linked to disease severity and persist in a high-risk population in Riverside County, California. *Frontiers in Public Health, 11*, 1129852.
13. Alisoltani, A., **Qiu, X.**, Jaroszewski, L., Sedova, M., Iyer, M., & Godzik, A. (2023). Gender differences in smoking-induced changes in the tumor immune microenvironment. *Archives of Biochemistry and Biophysics, 739*, 109579.
14. Bonenfant, J., Li, J., Nasouf, L., Miller, J., Lowe, T., Jaroszewski, L., **Qiu, X.**, ... Nair, M. G. (2022). Resistin concentration in early sepsis and all-cause mortality at a safety-net hospital in Riverside County. *Journal of Inflammation Research, 15*, 3925.
15. Zhou, W. L., Li, L. L., **Qiu, X. R.**, An, Q., & Li, M. H. (2017). Effects of combining insulin-like growth factor 1 and platelet-derived growth factor on osteogenesis around dental implants. *Chinese Journal of Dental Research, 20*(2), 105–109.
16. Li, M., Wang, Y., Zhou, H., Li, Y., **Qiu, X.**, Wang, X., & Zhao, Y. (2013). Inhibitory effects of mistletoe alkali on salivary adenoid cystic carcinoma cells. *Chemical Research in Chinese Universities, 29*(2), 275–279.
17. Li, M., Zhou, H., Wang, W., & **Qiu, X.** (2010). Inhibition of salidroside on salivary gland adenoid cystic carcinoma. *Chemical Research in Chinese Universities, 26*(6), 969–973.
18. Ruggiero-Ruff, R. E., et al. (2025). Single cell transcriptomic profiling of RELMα-regulated adipose macrophage and eosinophil subsets in diet-induced obesity. *The Journal of Immunology, 214*(Supplement_1), vkaf283-1099.
19. **Qiu, X.** (2025). Single-cell analysis of platelet heterogeneity in COVID-19 and sepsis using foundation models. *The Journal of Immunology, 214*(Supplement_1), vkaf283-692.
20. Takkouche, A., et al. (bioRxiv preprint, 2024). Divergence of the individual repeats in the leucine-rich repeat domains of human Toll-like receptors explain their diversity and functional adaptations. *bioRxiv*, 2024-09.

Manuscripts in Preparation / Under Review

1. **Qiu, X.** & Godzik, A. (in preparation, 2026; targeting AAAI 2027). The Oracle Gap: Benchmarking Unrealized Causal Information in Target Prioritization — a metric-agnostic benchmark (CausalTrapBench) and identification-limit sensitivity theory for causal target ranking. *AAAI Conference on Artificial Intelligence.*
2. **Qiu, X.** & Godzik, A. (under preparation, 2026). An out-of-distribution robustness benchmark of single-cell foundation models for in-hospital mortality prediction and cross-disease transfer from platelet transcriptomes. *Targeting Briefings in Bioinformatics.*
3. Ruggiero-Ruff, R. E.*, **Qiu, X.***, Vance, L., He, Y., Coss, D., Godzik, A.+, & Nair, M. G.+ (in revision, 2026). Single-cell heterogeneity of adipose tissue eosinophils is shaped by sex and RELMα and is linked to macrophage communication network in obesity. *co-first authors; +co-corresponding authors.
4. **Qiu, X.** & Godzik, A. (in preparation, 2026). An agentic target-validation system for therapeutic target hypotheses that flags its own contradictions — a six-layer propose-then-verify agent whose deterministic, constitution-governed cascading critic emits per-candidate `supported/partial/refuted/unknown` verdicts (TPR = TNR = 1.00 on a prospective FDA-target vs. housekeeping panel).

PhD Dissertation

Qiu, Xinru. Exploring Immune Cell Heterogeneity Through Single-Cell RNA Sequencing Analysis. University of California, Riverside, 2023.

Master's Thesis

Qiu, Xinru. pH-Sensitive Cytotoxicity of a Cell Penetrating Peptide Fused with a Histidine-Glutamate Co-oligopeptide. University of Southern California, 2015.

Presentations and Invited Talks

- Poster, RNA and AI Symposium 2024, San Diego, CA Oct 2024

  Single-Cell Analysis of Platelet Heterogeneity in COVID-19 and Sepsis Using Foundation Models
- Poster, La Jolla Immunology Conference, San Diego, CA Oct 2024

  Deciphering Abnormal Platelet Subpopulations in Inflammatory Diseases through Machine Learning and Single-Cell Transcriptomics
- Poster, RECOMB/ISCB Conference on Regulatory & Systems Genomics with DREAM Challenges, Los Angeles, CA Nov 2023

  Deciphering Abnormal Platelet Subpopulations in Inflammatory Diseases through Machine Learning and Single-Cell Transcriptomics
- Poster, Predictive Modeling in Biology & Medicine Conference, Riverside, CA Nov 2023

  Deciphering Abnormal Platelet Subpopulations in Inflammatory Diseases through Machine Learning and Single-Cell Transcriptomics
- Poster, Inaugural Symposium Center for RNA Biology, Riverside, CA Nov 2022

  Single-cell analysis identifies specific platelet subpopulations in COVID-19, sepsis, and systemic lupus erythematosus drive the disease severity
- Poster, 2022 UCR GGB Symposium, Riverside, CA Oct 2022

  Single-cell analysis identifies specific platelet subpopulations in COVID-19, sepsis, and systemic lupus erythematosus drive the disease severity
- Poster, 5th Annual Ultimate Biomed Retreat, Riverside, CA Oct 2022

  Single-cell analysis identifies specific platelet subpopulations in COVID-19, sepsis, and systemic lupus erythematosus drive the disease severity

- Poster, MolMed mini symposium, Riverside, CA May 2022
  Dynamic changes in human single-cell transcriptional signatures during fatal sepsis
- Poster, AAI IMMUNOLOGY2022, Portland, OR May 2022
  Dynamic changes in human single-cell transcriptional signatures during fatal sepsis
- Poster, 4th Annual Ultimate Biomed Retreat, Zoom Nov 2021
  Dynamic changes in human single-cell transcriptional signatures during fatal sepsis
- Talk, 2021 UCR GGB Symposium, Zoom Oct 2021
  Dynamic changes in human single-cell transcriptional signatures during fatal sepsis
- Talk and Poster, UC Irvine 2020 Immunology Fair, Zoom Dec 2020
  Analysis of Single Cell Expression Data from Sepsis Patients: A Pilot Study
- Talk, 3rd Annual Ultimate Biomed Retreat, Zoom Oct 2020
  Analysis of Single Cell Expression Data from Sepsis Patients: A Pilot Study
- Talk and Poster, 2020 UCR GGB Symposium, Zoom Oct 2020
  Analysis of Single Cell Expression Data from Sepsis Patients: A Pilot Study
- Talk and Poster, 2019 UCR GGB Symposium, Riverside, CA Sep 2019
  Mutations at protein-protein interfaces effect on drug resistance in cancer
- Poster, Inaugural MolMed Symposium, Riverside, CA Mar 2019
  Mutations at protein-protein interfaces effect on drug resistance in cancer

Reports by Media

- Report by news.ucr.edu `https://shorturl.at/brvL4`
  Article title: Protein found to protect females against obesity.
  Featured in UCR's Top Stories of 2023 `https://shorturl.at/kvzV2`
- Report by news.ucr.edu `https://shorturl.at/szNW`
  Article title: How sepsis need not be fatal.

Professional Affiliations

GGB Graduate Student Association, University of California, Riverside 2021 – 2022
Secretary

- Co-organized program symposiums and seminars, developing programs, inviting speakers, coordinating logistics, and ensuring smooth event execution.
- Organized and hosted Ph.D. candidate qualifying exam information seminar, developing content and structure, leveraging expertise to provide valuable insights about qualifying exam process.
- Worked with GSA President in facilitating monthly board meetings, coordinating with board members to ensure attendance, preparing meeting materials, and taking minutes to capture key decisions and action items.

Professional Service

Editorial Positions

- **Topic Editor**, *Frontiers in Bioinformatics* — Research Topic "Methods, Tools and Algorithms in Single Cell Bioinformatics, Volume II" (manuscript submission deadline February 2027). Scope spans AI and foundation models for predictive cell modeling; multimodal and spatial omics integration; atlas-scale computation, benchmarking, and clinical translation; and statistical inference, causal modeling, and interpretable AI.

Journal Reviews

Completed over 200 peer reviews for 130+ manuscripts across the following journals:

- Advanced Science (Wiley)
- Big Data and Cognitive Computing (MDPI)
- Biology (MDPI)
- BioMedInformatics (MDPI)
- Biomolecules (MDPI)
- BMC Bioinformatics (BMC/Springer)
- BMC Infectious Diseases (BMC/Springer)
- BMC Pharmacology and Toxicology (BMC/Springer)
- Brain Sciences (MDPI)
- Communications Biology (Nature)
- Data (MDPI)
- Frontiers in Endocrinology (Frontiers)
- Frontiers in Genetics (Frontiers)
- Frontiers in Pediatrics (Frontiers)
- Genes (MDPI)
- Genomics (Elsevier)
- Heliyon (Cell Press)
- Inflammation (Springer)
- International Journal of Environmental Research and Public Health (MDPI)
- International Journal of Molecular Sciences (MDPI)
- Journal of Clinical Medicine (MDPI)
- Journal of Molecular Biology (Elsevier)
- Journal of Personalized Medicine (MDPI)
- Mathematics (MDPI)
- Medicina (MDPI)
- Molecular Immunology (Elsevier)
- PeerJ (Taylor & Francis)
- Scientific Reports (Nature)
- Sensors (MDPI)

Conference and Workshop Reviews

- International Conference on Learning Representations (ICLR)
- International Conference on Machine Learning (ICML)
- GEM Workshop at ICLR 2024–2026
- AI4Science Workshop at ICML 2024, 2025

Grant Reviews

- UCR Minigrants for Undergraduate Research & Creative Activities

Professional Memberships

- **Member,** The International Society of Computational Biology (ISCB)
- **Member,** Human Cell Atlas (HCA)
- **Member,** The American Association of Immunologists (AAI)
- **Member (invited/nomination-based),** Sigma Xi (The Scientific Research Honor Society)

Languages

English: Fluent | **Chinese – Mandarin**: Native Language

Technical Skills and Tools

- **Programming Languages:** Python, R, Java, JavaScript, TypeScript, Node.js, HTML, SQL, Bash, Git
- **Operating Systems and Cloud Computing:** Unix, Linux, Windows, MacOS, Google Cloud Platform
- **High-Performance Computing (HPC):** Extensive experience with HPC clusters for large-scale computational tasks. SLURM batch job management and resource optimization. GPU-accelerated computing with CUDA for foundation model processing. Implemented parallel processing techniques and memory-efficient algorithms for analyses scaling to 1M+ cells (genome-wide Perturb-seq, ∼2M cells). Experienced in scaling bioinformatics workflows for HPC environments with containerization.
- **Container Technologies & Reproducible Research:** Singularity and Docker containerization for reproducible computational biology workflows. Version-controlled analysis pipelines. Container-based foundation model deployment on HPC systems. Established reproducible frameworks for large-scale biological studies.
- **Foundation Models for Single-Cell Biology:**
  - Models: STATE (2,058-dim), UCE (Universal Cell Embeddings, 1,280-dim), Geneformer (1,152-dim), scPRINT-2 (1,024-dim), scGPT (512-dim), TranscriptFormer, CellPLM
  - Large Language Models: C2S-Scale (27B parameter LLM for cell state annotation and biological interpretation)
  - Transfer Learning: Cross-species analysis, domain adaptation, embedding space optimization
  - Large-Scale Processing: 226K+ cell embedding generation, quality assessment frameworks
  - Disease Prediction: Foundation model-guided biomarker discovery, severity classification, therapeutic target prioritization
- **Machine Learning & Data Analysis Tools:**
  - Deep Learning Frameworks: TensorFlow, PyTorch, Keras, Hugging Face Transformers
  - Libraries: Pandas, NumPy, SciPy, Scikit-learn, Bioconductor, Tidyverse, RDKit
  - Models & Techniques: Classification and Regression (Linear & Logistic Regression, Random Forest, XGBoost, LightGBM, SVM, Naive Bayes, Decision Trees, Gradient Boosting, Elastic Net); Clustering (K-means, Hierarchical Clustering); Dimensionality Reduction (PCA, LDA, t-SNE, UMAP); Sequence Modeling (Hidden Markov Models, RNNs, LSTM); Deep Learning (CNNs, DNNs, Autoencoders, Diffusion Models, Transformers, Attention Mechanisms); Model Evaluation (ROC, AUC, Cross-Validation, SHAP, Hyperparameter Optimization); Ensemble Methods (Bagging, Boosting, Stacking)

  - Visualization Tools: ggplot2, Plotly, Seaborn, Matplotlib, Shiny, Tableau, BioRender
- **Advanced Single-Cell Analysis:**
  - Core Tools: CellRanger, Seurat, SCANPY, Monocle3, SingleR, scCATCH
  - RNA Velocity & Trajectory: scVelo, CellRank2, PAGA, MOSCOT, Waddington-OT for optimal transport-based lineage inference
  - Regulatory Networks: pySCENIC, GRNBoost2, dGCNA for transcription factor analysis and gene regulatory network inference
  - Cell Communication: CellChat for intercellular signaling and communication analysis
  - Quality Control: Comprehensive QC pipelines for large-scale datasets with automated validation
  - Embedding Analysis: Cosine similarity analysis, centroid-based comparisons, embedding quality assessment
- **Spatial Transcriptomics:**
  - Platforms: 10X Genomics Visium (including FFPE workflows)
  - Analysis Tools: Squidpy, Scanpy spatial modules
  - Methods: Spatial domain identification, niche analysis, ligand-receptor spatial mapping
  - Integration: Multi-modal integration of spatial and single-cell data
- **Bioinformatics Tools and Pipelines:**
  - Sequence Alignment: BLAST, Bowtie, BWA, STAR, HISAT2, Tophat2
  - Genomic Annotation: ANNOVAR, VEP, SnpEff
  - Variant Calling: GATK, VarScan, SAMtools, bcftools, Picard
  - Transcriptomics: Cufflinks, Kallisto, Salmon, DESeq2, edgeR, limma
  - Pathway Analysis: clusterProfiler, ReactomePA, GSVA
  - Perturbation Analysis: Large-scale gene perturbation screening (16,248+ targets), CRISPR screen analysis
  - Workflow Management: Nextflow, Snakemake, WDL/Cromwell
- **Protein Structure & Computational Drug Discovery:**
  - Structure Prediction: AlphaFold2, ESMFold, RoseTTAFold, OpenFold for protein structure prediction
  - Protein Design: ESM-2 (embeddings), ProteinMPNN, LigandMPNN, ProstT5, ProGen for protein engineering and generative AI-driven enzyme design
  - Generative Models: RFdiffusion for de novo protein backbone generation
  - Molecular Docking: DiffDock for ML-based ligand docking
  - Enzyme Engineering: Rational enzyme design, thermostability optimization, catalytic efficiency enhancement, structure-function relationships in PETase enzymes
  - Molecular Modeling: GROMACS for molecular dynamics simulations, peptide conformation prediction, protein-substrate binding analysis
  - Visualization: PyMOL, Chimera for structural analysis and visualization
  - Small Molecule Tools: RDKit for cheminformatics and drug-like compound analysis
- **Drug Repurposing & Target Discovery:**
  - Connectivity Mapping: CMap (Connectivity Map) for drug-disease signature matching
  - Pathway Enrichment: Enrichr for multi-library enrichment analysis
  - Disease-Gene Associations: DisGeNET for disease-gene relationship mining
  - Drug-Gene Interactions: DGIdb for druggable genome analysis
  - Target Prioritization: Multi-omic integration for therapeutic target identification and validation
- **Medicinal Chemistry & Drug Development:**
  - Drug Design: Rational drug design principles, peptide-based therapeutics, cell-penetrating peptides (CPPs)
  - ADMET Analysis: Cytotoxicity assays, cell permeability studies, pH-dependent activity profiling
  - Pharmacokinetics: Drug metabolism considerations, structure-activity relationships, therapeutic optimization
  - Drug Delivery Systems: Targeted drug delivery, cellular uptake mechanisms, pH-sensitive therapeutics
- **Computational Efficiency Optimization:**

- HVG-based gene selection achieving 92% reduction with biological signal preservation
  - Batch processing optimization for HPC environments (188-chunk parallel processing)
  - Memory-efficient algorithms for large-scale biological data (1M+ cells)
  - GPU-accelerated computation for foundation model inference with 5-10x speedup
- **Clinical Genomics & Pharmacogenomics:**
  - Variant Classification: ACMG guidelines implementation for clinical variant interpretation
  - Pharmacogenomics Algorithms: Star allele calling (CYP2D6, CYP2C19), CNV detection, haplotype phasing, diplotype inference
  - Knowledge Bases: PharmVar, PharmGKB, CPIC database integration, parsing, and cross-validation
  - Genotype-Phenotype Translation: Clinical evidence level assessment, CPIC guideline implementation for decision support
  - Clinical Reporting: DRAGEN Bio-IT platform, automated report generation
  - NGS Quality Control: Pipeline validation and quality assurance for clinical applications
- **Databases:**
  - Genomic: 1000 Genomes Browser, dbSNP, gnomAD, HGMD, COSMIC, ClinVar
  - Transcriptomic: GEO, SRA, ENCODE, ArrayExpress
  - Pathway & Protein: KEGG, Reactome, IPA, Gene Ontology (GO), PANTHER, SWISS-PROT, PDB, STRING, UniProt
  - Genome Browsers: UCSC Genome Browser, Ensembl, IGV (Integrative Genomics Viewer)
  - Pharmacogenomics: PharmVar, PharmGKB, CPIC
  - Drug Discovery: CMap, DGIdb, DisGeNET, DrugBank